VolunTier - Detailed Test Case Descriptions

Team 3 - DIGT2107. FR-001 to FR-009 match functional requirements 1 to 9 in section 2.1 of the Requirements document. NFR-001 to NFR-007 follow the order of the non-functional requirements in section 2.2: Performance, Usability, Reliability, Security, Scalability, Validation, Maintainability.

TC-001

Test Case ID	TC-001
Related Requirement(s)	FR-001
Title	Login - Valid Volunteer Credentials
Preconditions	A volunteer account exists. The user is on the Sign In screen.
Test Steps	1. Enter the volunteer email and password 2. Click "Sign In"
Expected Result	The volunteer dashboard opens. Only volunteer features show.
Actual Result	As expected. Automated test TC001 passed.
Status	Pass
Priority	High

TC-002

Test Case ID	TC-002
Related Requirement(s)	FR-001
Title	Login - Valid Coordinator Credentials
Preconditions	A coordinator account exists. The user is on the Sign In screen.
Test Steps	1. Enter the coordinator email and password 2. Click "Sign In"
Expected Result	The coordinator dashboard opens with opportunity management features.
Actual Result	As expected. Automated test TC002 passed.
Status	Pass
Priority	High

TC-003

Test Case ID	TC-003
Related Requirement(s)	FR-001
Title	Login - Valid Administrator Credentials
Preconditions	An administrator account exists. The user is on the Sign In screen.
Test Steps	1. Enter the administrator email and password 2. Click "Sign In"
Expected Result	The administrator dashboard opens with user management and program summaries.
Actual Result	As expected. Automated test TC003 passed.
Status	Pass
Priority	High

TC-004

Test Case ID	TC-004
Related Requirement(s)	FR-001
Title	Login - Invalid Password
Preconditions	A volunteer account exists. The user is on the Sign In screen.
Test Steps	1. Enter a valid email and a wrong password 2. Click "Sign In"
Expected Result	Login fails. A clear error message shows. The user stays on the Sign In screen.
Actual Result	As expected. Automated test TC004 passed.
Status	Pass
Priority	High

TC-005

Test Case ID	TC-005
Related Requirement(s)	FR-001, NFR-006
Title	Login - Empty Required Fields
Preconditions	The user is on the Sign In screen.
Test Steps	1. Leave email and password empty 2. Click "Sign In"
Expected Result	The system marks the required fields and shows an error. No login occurs.
Actual Result	As expected. Automated test TC005 passed.
Status	Pass
Priority	Medium

TC-006

Test Case ID	TC-006
Related Requirement(s)	FR-001, NFR-004
Title	Role Restriction - Volunteer Cannot Access Coordinator Features
Preconditions	The user is logged in as a Volunteer.
Test Steps	1. Look for "Post an Opportunity" and "Attendance & Hours" in the navigation 2. Try to open the coordinator page directly (URL or menu)
Expected Result	The features are hidden or access is denied. No coordinator data shows.
Actual Result	As expected. Automated test TC006 passed.
Status	Pass
Priority	High

TC-007

Test Case ID	TC-007
Related Requirement(s)	FR-001, NFR-004
Title	Role Restriction - Coordinator Cannot Access Administrator Features
Preconditions	The user is logged in as a Coordinator.
Test Steps	1. Look for "Manage Coordinator Access" and "Program Overview" in the navigation 2. Try to open the administrator page directly
Expected Result	Access is denied. No administrator data shows.
Actual Result	As expected. Automated test TC007 passed.
Status	Pass
Priority	High

TC-008

Test Case ID	TC-008
Related Requirement(s)	FR-002
Title	Browse Opportunities - List Displays Details
Preconditions	The user is logged in as a Volunteer. At least three opportunities exist.
Test Steps	1. Open the Browse Opportunities page
Expected Result	Each opportunity shows date, location, description, and remaining spaces.
Actual Result	As expected. Automated test TC008 passed.
Status	Pass
Priority	High

TC-009

Test Case ID	TC-009
Related Requirement(s)	FR-002
Title	Search Opportunities - Matching Term
Preconditions	The user is logged in as a Volunteer. An opportunity titled "Food Bank Sorting" exists.
Test Steps	1. Open Browse Opportunities 2. Enter "Food Bank" in the search field
Expected Result	Only matching opportunities show, including "Food Bank Sorting".
Actual Result	As expected. Automated test TC009 passed.
Status	Pass
Priority	High

TC-010

Test Case ID	TC-010
Related Requirement(s)	FR-002
Title	Search Opportunities - No Matches
Preconditions	The user is logged in as a Volunteer.
Test Steps	1. Open Browse Opportunities 2. Enter a term that matches nothing (e.g., "zzzzqq")
Expected Result	A "no matches" message shows. The user can change the search or filters.
Actual Result	As expected. Automated test TC010 passed.
Status	Pass
Priority	Medium

TC-011

Test Case ID	TC-011
Related Requirement(s)	FR-002
Title	Filter Opportunities - By Category or Date
Preconditions	The user is logged in as a Volunteer. Opportunities exist in more than one category or date.
Test Steps	1. Open Browse Opportunities 2. Select a filter value
Expected Result	The list shows only opportunities that match the filter.
Actual Result	As expected. Automated test TC011 passed.
Status	Pass
Priority	Medium

TC-012

Test Case ID	TC-012
Related Requirement(s)	FR-002
Title	View Opportunity Details
Preconditions	The user is logged in as a Volunteer. Opportunities exist.
Test Steps	1. Open Browse Opportunities 2. Select one opportunity
Expected Result	The Opportunity Details page shows title, description, date, time, location, capacity, and a Register button.
Actual Result	As expected. Automated test TC012 passed.
Status	Pass
Priority	Medium

TC-013

Test Case ID	TC-013
Related Requirement(s)	FR-003
Title	Register for Opportunity - Space Available
Preconditions	Logged in as a Volunteer. The opportunity has open spaces. The volunteer is not registered.
Test Steps	1. Open the opportunity details 2. Click "Register"
Expected Result	A confirmation shows. The opportunity appears in Upcoming Registrations. Remaining spaces fall by one.
Actual Result	As expected. Automated test TC013 passed.
Status	Pass
Priority	High

TC-014

Test Case ID	TC-014
Related Requirement(s)	FR-003
Title	Register for Opportunity - Duplicate Registration
Preconditions	Logged in as a Volunteer. The volunteer is already registered for the opportunity.
Test Steps	1. Open the same opportunity details 2. Click "Register"
Expected Result	The system refuses and explains that the volunteer is already registered. No second record is created.
Actual Result	As expected. Automated test TC014 passed.
Status	Pass
Priority	High

TC-015

Test Case ID	TC-015
Related Requirement(s)	FR-003
Title	Register for Opportunity - Capacity Reached
Preconditions	Logged in as a Volunteer. The opportunity has 0 remaining spaces.
Test Steps	1. Open the opportunity details 2. Click "Register"
Expected Result	The system refuses and explains that the opportunity is full. No registration is saved.
Actual Result	As expected. Automated test TC015 passed.
Status	Pass
Priority	High

TC-016

Test Case ID	TC-016
Related Requirement(s)	FR-003
Title	Register for Opportunity - Last Remaining Space
Preconditions	Logged in as a Volunteer. The opportunity has exactly 1 remaining space.
Test Steps	1. Register for the opportunity 2. Log in as a second volunteer and try to register
Expected Result	The first volunteer registers. Remaining spaces show 0. The second volunteer is refused.
Actual Result	As expected. Automated test TC016 passed.
Status	Pass
Priority	Medium

TC-017

Test Case ID	TC-017
Related Requirement(s)	FR-004
Title	Withdraw - More Than Three Days Before Event
Preconditions	Logged in as a Volunteer. The volunteer is registered for an event 7 days away.
Test Steps	1. Open My Volunteering 2. Select the opportunity and click "Withdraw" 3. Confirm
Expected Result	The registration is removed. A confirmation shows. The space is available again.
Actual Result	As expected. Automated test TC017 passed.
Status	Pass
Priority	Medium

TC-018

Test Case ID	TC-018
Related Requirement(s)	FR-004
Title	Withdraw - Within Three Days of Event
Preconditions	Logged in as a Volunteer. The volunteer is registered for an event 2 days away.
Test Steps	1. Open My Volunteering 2. Select the opportunity and click "Withdraw"
Expected Result	Withdrawal is blocked. The message explains the three-day cutoff. The registration stays.
Actual Result	As expected. Automated test TC018 passed.
Status	Pass
Priority	Medium

TC-019

Test Case ID	TC-019
Related Requirement(s)	FR-004
Title	Withdraw - Exactly Three Days Boundary
Preconditions	Logged in as a Volunteer. The volunteer is registered for an event exactly 72 hours away.
Test Steps	1. Open My Volunteering 2. Select the opportunity and click "Withdraw"
Expected Result	Withdrawal is blocked because the event is not more than three days away.
Actual Result	As expected. Automated test TC019 passed.
Status	Pass
Priority	Low

TC-020

Test Case ID	TC-020
Related Requirement(s)	FR-005
Title	My Volunteering - Upcoming and History
Preconditions	Logged in as a Volunteer. The volunteer has upcoming and past registrations.
Test Steps	1. Open My Volunteering
Expected Result	Upcoming registrations and past participation show in separate, clear sections.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-021

Test Case ID	TC-021
Related Requirement(s)	FR-005, FR-007
Title	My Volunteering - Hours Show After Coordinator Records Them
Preconditions	A coordinator recorded 3 hours for the volunteer. Logged in as that Volunteer.
Test Steps	1. Open My Volunteering 2. View participation history
Expected Result	The history shows the opportunity as attended with 3 completed hours. The total hours include 3.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-022

Test Case ID	TC-022
Related Requirement(s)	FR-005
Title	My Volunteering - No Records
Preconditions	Logged in as a new Volunteer with no registrations.
Test Steps	1. Open My Volunteering
Expected Result	A helpful message shows that there are no records.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Low

TC-023

Test Case ID	TC-023
Related Requirement(s)	FR-006
Title	Create Opportunity - Valid Inputs
Preconditions	Logged in as a Coordinator.
Test Steps	1. Open Post an Opportunity 2. Enter title, description, future date, time, location, and capacity 10 3. Click "Post"
Expected Result	The opportunity saves. It appears in the list that volunteers see.
Actual Result	As expected. Automated test TC023 passed.
Status	Pass
Priority	High

TC-024

Test Case ID	TC-024
Related Requirement(s)	FR-006, NFR-006
Title	Create Opportunity - Missing Required Fields
Preconditions	Logged in as a Coordinator.
Test Steps	1. Open Post an Opportunity 2. Leave the title empty and fill other fields 3. Click "Post"
Expected Result	The system explains the missing field. Nothing is saved.
Actual Result	As expected. Automated test TC024 passed.
Status	Pass
Priority	High

TC-025

Test Case ID	TC-025
Related Requirement(s)	FR-006, NFR-006
Title	Create Opportunity - Invalid Date or Capacity
Preconditions	Logged in as a Coordinator.
Test Steps	1. Open Post an Opportunity 2. Enter a past date or a capacity of 0 or a negative number 3. Click "Post"
Expected Result	The system rejects the input with a clear message. Nothing is saved.
Actual Result	As expected. Automated test TC025 passed.
Status	Pass
Priority	Medium

TC-026

Test Case ID	TC-026
Related Requirement(s)	FR-006
Title	Edit Opportunity - Valid Change
Preconditions	Logged in as a Coordinator. An opportunity exists.
Test Steps	1. Open the opportunity for editing 2. Change the description 3. Click "Save"
Expected Result	The change saves and shows in the opportunity list.
Actual Result	As expected. Automated test TC026 passed.
Status	Pass
Priority	Medium

TC-027

Test Case ID	TC-027
Related Requirement(s)	FR-006, FR-009
Title	Cancel Opportunity - Registered Volunteers Notified
Preconditions	Logged in as a Coordinator. An opportunity has two registered volunteers.
Test Steps	1. Open the opportunity 2. Click "Cancel Opportunity" and confirm
Expected Result	The status changes to cancelled. Both volunteers receive a notification that names the opportunity.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	High

TC-028

Test Case ID	TC-028
Related Requirement(s)	FR-007
Title	View Registrations for an Opportunity
Preconditions	Logged in as a Coordinator. An opportunity has registrations.
Test Steps	1. Select the opportunity 2. Open its registration list
Expected Result	The registered volunteers and the number of available places show. No data changes.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-029

Test Case ID	TC-029
Related Requirement(s)	FR-007
Title	Record Attendance and Hours - Valid
Preconditions	Logged in as a Coordinator. The event date has passed. Volunteers are registered.
Test Steps	1. Open the registration list 2. Mark one volunteer as attended 3. Enter 3 hours 4. Click "Save"
Expected Result	Attendance and hours save. They appear in the volunteer's participation history.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-030

Test Case ID	TC-030
Related Requirement(s)	FR-007, NFR-006
Title	Record Hours - Invalid Hours
Preconditions	Logged in as a Coordinator. The event date has passed.
Test Steps	1. Open the registration list 2. Mark a volunteer as attended 3. Enter -2, or a letter, or leave empty 4. Click "Save"
Expected Result	The system asks for a correction. Nothing is saved.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-031

Test Case ID	TC-031
Related Requirement(s)	FR-008
Title	Grant Coordinator Access
Preconditions	Logged in as an Administrator. A volunteer account exists.
Test Steps	1. Open the user management page 2. Select the volunteer 3. Grant coordinator access and save
Expected Result	The change saves. After the next login, the user sees coordinator features.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-032

Test Case ID	TC-032
Related Requirement(s)	FR-008
Title	Remove Coordinator Access
Preconditions	Logged in as an Administrator. A coordinator account exists.
Test Steps	1. Open the user management page 2. Select the coordinator 3. Remove coordinator access and save
Expected Result	The change saves. The user no longer sees coordinator features.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-033

Test Case ID	TC-033
Related Requirement(s)	FR-008
Title	Review Program Summaries
Preconditions	Logged in as an Administrator. Opportunities, registrations, and hours exist.
Test Steps	1. Open the Program Overview page
Expected Result	The summary shows counts of opportunities, registrations, attendance, and total hours. No data changes.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-034

Test Case ID	TC-034
Related Requirement(s)	FR-008
Title	Program Summaries - No Data For Period
Preconditions	Logged in as an Administrator. No data exists for the selected period.
Test Steps	1. Open Program Overview 2. Select a period with no activity
Expected Result	A clear "no information" message shows.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Low

TC-035

Test Case ID	TC-035
Related Requirement(s)	FR-009
Title	Notification - Opportunity Time or Location Changed
Preconditions	A volunteer is registered. Logged in as the Coordinator.
Test Steps	1. Edit the opportunity time or location 2. Save 3. Log in as the registered volunteer and open notifications
Expected Result	The volunteer has a notification that names the opportunity and states what changed.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-036

Test Case ID	TC-036
Related Requirement(s)	FR-009
Title	Notification - Unregistered Volunteer Not Notified
Preconditions	A second volunteer is not registered for the opportunity.
Test Steps	1. Edit the opportunity date and save 2. Log in as the unregistered volunteer
Expected Result	The unregistered volunteer receives no notification.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Low

TC-037

Test Case ID	TC-037
Related Requirement(s)	NFR-001
Title	Performance - Lists Load Within Two Seconds
Preconditions	Database holds hundreds of opportunities and thousands of records.
Test Steps	1. Open Browse Opportunities 2. Open My Volunteering 3. Measure the time to display each page
Expected Result	Each page displays within two seconds.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-038

Test Case ID	TC-038
Related Requirement(s)	NFR-003
Title	Reliability - No Duplicate Records On Rapid Double Click
Preconditions	Logged in as a Volunteer. The opportunity has open spaces.
Test Steps	1. Double-click "Register" quickly 2. Open My Volunteering
Expected Result	Only one registration exists. Remaining spaces fall by one only.
Actual Result	As expected. Automated test TC038 passed.
Status	Pass
Priority	High